

Connor Savugot

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1 Technical Skills

- **Languages:** C, C++, Python, Bash, MATLAB, VBA
- **Embedded Systems:** FreeRTOS, Embedded Linux, systemd services, IPMI, external watchdogs, BMS integration, Yocto/Arago, device trees, board bring-up
- **Platforms:** TI MSP430, TI TMS320, TI AM62-class / ARM Linux SoCs, STM32, dsPIC
- **Protocols & Peripherals:** CAN, CAN FD, UART, RS232, SPI, I2C, ADC, PWM, GPIO, timers, JTAG
- **Hardware & Tools:** Through-hole and SMT soldering, PCB rework, breadboard prototyping, KiCad, Git/GitLab, JTAG, oscilloscopes, logic analyzers

2 Professional Experience

AEGIS Power Systems

Current

Embedded Firmware Engineer

- Developed firmware control for custom bidirectional buck-boost power supplies using 24 independently configurable PWM channels across transformer-coupled input and output stages.
- Built a centralized configuration framework that translated requested input and output voltages into coordinated PWM timing, duty-cycle, and switching behavior across paired converter channels; integrated ADC feedback and safe-operating constraints.
- Evaluated BMS, monitoring, interface, and support ICs against electrical limits, host protocols, protection features, footprints, tool support, and sourcing risk; converted datasheet and reference-design findings into firmware and integration plans.
- Root-caused I2C, CAN/CAN FD, SPI, and UART faults by correlating register transactions and analyzer captures with schematics, timing requirements, bus configuration, and known-good hardware; implemented focused tests for peripheral-by-peripheral bring-up.
- Characterized IPMI 2.0 and chassis-management services on embedded power hardware, validating IPMB addressing plus sensor and FRU access while separating transport, session, authentication, and firmware-compatibility failures.
- Created external-watchdog and systemd services, device-tree and Yocto/Arago support, and Python/Bash tooling for reproducible monitoring, flashing, automated validation, and field diagnostics on TI AM62-class Linux systems.

AEGIS Power Systems

May 2024 – Aug 2024

Computer Science Intern

- Built a secured chroot environment and restricted SSH command-line interface on a stripped TI Yocto/Arago system, using custom Bash/Python commands to inspect and control embedded power hardware.
- Created Python and Bash firmware flashing utilities with packetized transfer, checksum validation, and SPI reflashing support for TI TMS320 targets.
- Reverse-engineered serial command transactions from limited vendor documentation, generated checksums, and helped construct a relay, resistive-load, and power-supply test circuit for burn-in monitoring.
- Authored Git and embedded-system guides to standardize firmware delivery, testing, and toolchain onboarding.

TEAM Industries

Jun 2022 – Aug 2022

Engineering Intern

- Automated Excel-based quality-control reporting and revision tracking using VBA macros and structured technical worksheets.
- Produced technical documentation that clarified engineering changes and supported manufacturing process reviews.

3 Selected Technical Projects

EV Active Sensor Adapter

Senior Design, MSP430FR2355

- Defined the end-to-end architecture and interface specifications for an EV active-sensor adapter integrating an MSP430, BMS and power-management hardware, sensors, custom PCBs, embedded firmware, and system telemetry.
- Evaluated BMS and supporting components against electrical, sensing, protection, communication, packaging, and sourcing requirements; documented trade studies and translated selected-device requirements into hardware and firmware interfaces.
- Prototyped interrupt-driven SPI master/slave register transactions and integrated ADC acquisition, UART diagnostics, timers, GPIO, and LCD telemetry while validating each subsystem before full-system integration.
- Maintained requirements, architecture diagrams, interface definitions, test plans, design decisions, and progress reports; coordinated work across team members and communicated specifications, risks, and results during design reviews.

4 Education

North Carolina State University

May 2026

B.S. Computer Engineering

5 Technical Leadership

Lead Instructor, Project-Based Electronics Workshop

Two Semesters

- Co-led a project-based class from concept through final demonstration, guiding students as they scoped, designed, debugged, and completed original electronics projects.
- Developed and taught through-hole and SMT soldering theory, safety, and hands-on technique using two custom NCSU IEEE flashing-light PCB keepsakes: one through-hole and one surface-mount.

Embedded Systems Teaching Assistant

NC State University

- Delivered supplemental lectures and test preparation; taught embedded theory, code debugging, lab techniques, and systematic hardware bring-up while advising students on projects, coursework, and ECE pathways.
- Used question-driven, one-on-one coaching to help students diagnose their own problems and form lasting professional relationships, raising the class average from its typical C+ to an A-.